

NextHikes IT Solutions
Bike Sharing Demand Analysis Project
Using Excel

Preprocess Summary Report

Project 1 - Data Set 1 Preprocess Report

Preprocessing Summary Report	
Step	Detail
Original Shape	610 rows $\times$ 10 columns
After Preprocessing	610 rows $\times$ 11 columns
Duplicates Found	0 (None removed)
Missing Values	0 in all columns

Transformations Applied	
season	1→Spring, 2→Summer, 3→Fall, 4→Winter
yr	0→2011, 1→2012
weathersit	1→Clear, 2→Mist/Cloudy, 3→Light Rain/Snow, 4→Heavy Rain/Snow
holiday	True/False → Yes/No
weekday	0→Sunday ... 6→Saturday
mnth	1→Jan ... 12→Dec
temp_celsius (new col)	Temp_Normalized × 41 → Actual °C
dteday	Datetime → Date only (YYYY-MM-DD)

Transformations Applied	
season	1→Spring, 2→Summer, 3→Fall, 4→Winter
yr	0→2011, 1→2012
weathersit	1→Clear, 2→Mist/Cloudy, 3→Light Rain/Snow, 4→Heavy Rain/Snow
holiday	True/False → Yes/No
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mnth	1→Jan ... 12→Dec
temp_celsius (new col)	Temp_Normalized × 41 → Actual °C
dteday	Datetime → Date only (YYYY-MM-DD)

Column Renames

instant → Record_ID

dteday → Date

season → Season

yr → Year

mnth → Month

hr → Hour

holiday → Is_Holiday

weekday → Weekday

weathersit → Weather_Situation

temp → Temp_Normalized

Project 2 - Data Set 2 Preprocess Report

OVERVIEW

Original Shape	610 rows $\times$ 15 columns
Final Shape	610 rows $\times$ 10 columns

STEP 1 — Junk Column Removal

Columns Removed	8 unwanted columns dropped
'Unnamed: 0'	Redundant row index copy
'Unnamed: 8/12'	Completely empty columns (610/610 NaN)
'Mean','Mode','Median'	Completely empty columns (610/610 NaN)
'0.2879' + 'Unnamed: 14'	Stats sidebar accidentally embedded in sheet

STEP 2 — Duplicate Removal

Duplicates Found & Dropped	0
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STEP 3 — Missing Value Treatment

'atemp' column	11 nulls filled with median = 0.197
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Strategy	Median imputation (robust to skew/outliers)
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STEP 4 — Numeric Inconsistency Fix

'hum' = 1.0 (100% humidity)	1 row(s) capped to 0.99 (sensor anomaly)
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STEP 5 — Denormalization (New Columns)

'atemp_celsius'	$\text{Feels_Like_Temp_Norm} \times 50 \rightarrow ^\circ\text{C}$
'humidity_pct'	$\text{Humidity_Norm} \times 100 \rightarrow \%$
'windspeed_kmh'	$\text{Windspeed_Norm} \times 67 \rightarrow \text{km/h}$

STEP 6 — Column Renames

instant $\rightarrow$ Record_ID	atemp $\rightarrow$ Feels_Like_Temp_Norm
hum $\rightarrow$ Humidity_Norm	windspeed $\rightarrow$ Windspeed_Norm
casual $\rightarrow$ Casual_Users	registered $\rightarrow$ Registered_Users
cnt $\rightarrow$ Total_Users	

Project 3 - Data Set 3 Preprocess Report

DATASET OVERVIEW

Original Shape	390 rows × 16 columns
Final Shape	390 rows × 17 columns
Date Range	2011-01-28 to 2011-02-14 (18 days)
Record_ID Range	611 – 1000 (continuation of datasets 1 & 2)

STEP 1 — Duplicate Removal

Duplicates Found & Dropped	0
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STEP 2 — Missing Values

Missing Values Found	0 (dataset structurally complete)
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STEP 3 — Zero-Variance Column Drop

'season' dropped	All 390 rows = 1 (Spring only) — no analytical value
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'yr' dropped	All 390 rows = 0 (year 2011 only) — no analytical value
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'holiday' dropped	All 390 rows = False — no variation in this slice
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STEP 4 — Humidity Anomaly Fix

hum = 1.0 rows found

15 rows across 4 dates (sensor saturation)

Fix Strategy

Each anomalous value → daily median of valid (non-1.0) hum for that date

Anomaly Details

See 'Anomaly Log' sheet for full record-level breakdown

STEP 5 — Valid Edge Cases (Retained)

windspeed = 0

75 rows retained — calm-wind hours are valid observations

Missing hours per day

Partial-day recordings retained as-is (reflect actual sensor gaps)

STEP 6 — Categorical Decoding

mnth	1–12 → Jan – Dec
weathersit	1→Clear, 2→Mist/Cloudy, 3→Light Rain/Snow, 4→Heavy Rain/Snow
weekday	0→Sunday ... 6→Saturday

STEP 7 — Denormalization (4 New Columns)

temp_celsius	$\text{Temp_Norm} \times 41 \rightarrow \text{Actual temperature } ^\circ\text{C}$
atemp_celsius	$\text{FeelsLike_Norm} \times 50 \rightarrow \text{Feels-like temperature } ^\circ\text{C}$
humidity_pct	$\text{Humidity_Norm} \times 100 \rightarrow \text{Relative humidity } \%$
windspeed_kmh	$\text{Windspeed_Norm} \times 67 \rightarrow \text{Wind speed km/h}$

STEP 8 — Column Renames & Reorder

instant → Record_ID

dteday → Date

mnth → Month

hr → Hour

weekday → Weekday

weathersit → Weather_Situation

temp → Temp_Norm

atemp → FeelsLike_Norm

hum → Humidity_Norm

windspeed → Windspeed_Norm

casual → Casual_Users

registered → Registered_Users

cnt → Total_Users

COLUMN GLOSSARY

Record_ID	Unique hourly record (611–1000, continuation of datasets 1 & 2)
Date	Observation date (YYYY-MM-DD)
Month / Hour / Weekday	Temporal labels
Weather_Situation	Clear / Mist/Cloudy / Light Rain/Snow / Heavy Rain/Snow
Temp_Norm / FeelsLike_Norm	Normalized values (0–1 scale)
temp_celsius / atemp_celsius	Human-readable temperature in °C
humidity_pct / windspeed_kmh	Humidity % and Wind speed km/h
Casual / Registered / Total_Users	Bike rental user counts per hour

Data Set A (Merge Project 1+2)

MERGE DETAILS

Merge Key	Record_ID (inner join)
Dataset 1 Shape	610 rows × 11 columns
Dataset 2 Shape	610 rows × 10 columns (2 header rows fixed)
After Merge	610 rows × 20 columns
Final Shape	610 rows × 16 columns

STEP 1 — Duplicate Removal

Duplicates Found	0
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STEP 2 — Missing Value Treatment

Missing Values Found	0 (none after merge)
Strategy	Median imputation for numeric columns

STEP 3 — Redundant Column Removal

'Temp_Normalized'	Kept Temp_Celsius (human-readable °C)
'Feels_Like_Temp_Norm'	Kept atemp_celsius → renamed FeelsLike_Temp_C
'Humidity_Norm'	Kept Humidity_Pct (% form is clearer)
'Windspeed_Norm'	Kept windspeed after recomputing km/h
'windspeed_kmh' (old)	Dropped — had integer truncation error; recomputed as float

STEP 4 — Numeric Fix

Windspeed_kmh	Recomputed: Windspeed_Norm $\times$ 67 $\rightarrow$ float (was int-truncated)
Date column	Datetime stripped to Date only (YYYY-MM-DD)

STEP 5 — Column Reordering & Rename

Final Column Order	ID $\rightarrow$ Date/Time $\rightarrow$ Day Info $\rightarrow$ Weather $\rightarrow$ Climate $\rightarrow$ Users
atemp_celsius	$\rightarrow$ FeelsLike_Temp_C
humidity_pct	$\rightarrow$ Humidity_Pct

FINAL COLUMN GLOSSARY

Record_ID	Unique hourly record identifier
Date	Date of observation (YYYY-MM-DD)
Year / Month / Season	Temporal labels (2011/2012, Jan–Dec, Spring–Winter)
Hour	Hour of day (0–23)
Weekday	Day name (Sunday–Saturday)
Is_Holiday	Yes / No
Weather_Situation	Clear / Mist/Cloudy / Light Rain/Snow / Heavy Rain/Snow
Temp_Celsius	Actual temperature in °C
FeelsLike_Temp_C	Feels-like temperature in °C
Humidity_Pct	Relative humidity in %
Windspeed_kmh	Wind speed in km/h
Casual_Users	Non-registered bike rental users
Registered_Users	Registered bike rental users
Total_Users	Total rentals (Casual + Registered)

Data Set B (Merge Data Set A + Project 3)

SOURCE FILES

File 1	Data_Set_A.xlsx → 610 rows
File 2	Project_3.xlsx → 390 rows
Merge Strategy	Vertical concat (no overlapping IDs)

SHAPE

After Merge	1,000 rows × 16 columns
After Preprocessing	1,000 rows × 16 columns

MISSING VALUES

Missing Before

0

Strategy

Median imputation (numeric), "No"/"Clear" (categorical)

Missing After

0

DUPLICATES

Duplicates Found

0

Duplicates Dropped

0

STRING / CATEGORICAL FIXES

Columns Cleaned	Month, Season, Weekday, Is_Holiday, Weather_Situation
Operations	strip whitespace, title-case, variant mapping
Weather Variants Unified	4 categories standardised

NUMERIC CONSISTENCY FIXES

Humidity_Pct	Clipped to valid range 0–100
User Counts	Negative values clipped to 0
Total_Users	Recomputed as Casual + Registered

NEW COLUMNS ADDED (from P3)

Year	Extracted from Date
Season	Derived from Month (Spring/Summer/Autumn/Winter)
Is_Holiday	Defaulted to "No" for P3 subset